

AI Smart Shopping Cart Project

1. Project Title

AI Based Smart Shopping Cart using RFID, Computer Vision, and IoT

2. Aim of the Project

To develop an intelligent shopping cart system that automatically detects products placed in the cart, calculates the total bill, displays the product details, and reduces billing queue time using AI and IoT technologies.

3. Objectives

- Automatic product identification
 - Real-time bill calculation
 - Reduce waiting time at billing counter
 - Smart theft detection
 - Cashless and faster shopping
 - User-friendly shopping experience
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4. Features of the Project

Basic Features

- Automatic product scanning
- LCD/OLED display for total bill
- Product addition and removal
- RFID-based identification
- Wireless bill update

AI Features

- Object recognition using camera
- Voice assistant support
- Customer recommendation system
- Theft detection using AI
- Smart inventory management

5. Block Diagram

[Insert AI Smart Shopping Cart Block Diagram Here]

The block diagram includes:

- RFID Tag and RFID Reader
- ESP32 Microcontroller
- OLED Display
- Load Cell with HX711
- ESP32-CAM for AI Object Detection
- Buzzer Alert System
- WiFi/Firebase Cloud Connection
- Mobile App/Web Dashboard
- Power Supply Unit

5.1 Block Diagram Explanation

Main Blocks

1. ESP32 Microcontroller
2. RFID Reader
3. RFID Tags
4. OLED/LCD Display
5. Load Cell Sensor
6. Camera Module
7. Buzzer
8. WiFi Communication
9. Firebase/Cloud Database
10. Mobile Application/Web Dashboard

6. Working Principle

1. Customer places product into the cart.
2. RFID reader scans the RFID tag attached to product.
3. ESP32 identifies product and fetches details.
4. Product name and price displayed on OLED.
5. Total bill updated automatically.
6. AI camera verifies product using object detection.
7. Load cell checks weight mismatch for theft prevention.
8. Final bill sent to mobile app or cloud database.

7. Hardware Components Required

Component	Quantity	Purpose
ESP32 Dev Board	1	Main controller
RFID Reader RC522	1	Product identification
RFID Tags	Multiple	Product tagging
OLED Display 0.96 inch	1	Display details
HX711 Load Cell Amplifier	1	Weight sensing
Load Cell 5kg	1	Product weight measurement
ESP32-CAM	1	AI object detection
Buzzer	1	Alerts
Push Buttons	2	Remove/reset item
Li-ion Battery	1	Power supply
Jumper Wires	Multiple	Connections
Breadboard/PCB	1	Circuit assembly
Power Bank	1	Portable power

8. Software Requirements

Software	Purpose
Arduino IDE	ESP32 programming
Python	AI object detection
OpenCV	Computer vision
TensorFlow/YOLO	AI model
Firebase	Cloud database
Blynk App	Mobile monitoring

9. Specifications

ESP32 Specifications

- Operating Voltage: 3.3V
- WiFi and Bluetooth inbuilt
- Dual-core processor
- Clock Frequency: 240MHz
- GPIO Pins: 30+

RFID RC522 Specifications

- Frequency: 13.56 MHz
- Communication: SPI
- Read Distance: 2–5 cm
- Supply Voltage: 3.3V

OLED Display Specifications

- Resolution: 128x64
- Communication: I2C
- Voltage: 3.3V–5V

Load Cell Specifications

- Capacity: 5kg
- Accuracy: High precision
- Amplifier: HX711

ESP32-CAM Specifications

- OV2640 camera
- WiFi enabled
- Supports AI processing

10. Circuit Connections

RFID RC522 to ESP32

RFID Pin	ESP32 Pin
SDA	GPIO5

RFID Pin	ESP32 Pin
SCK	GPIO18
MOSI	GPIO23
MISO	GPIO19
RST	GPIO22
GND	GND
3.3V	3.3V

OLED Display to ESP32

OLED Pin	ESP32 Pin
VCC	3.3V
GND	GND
SCL	GPIO22
SDA	GPIO21

HX711 Load Cell Connections

HX711 Pin	ESP32 Pin
DT	GPIO4
SCK	GPIO2
VCC	3.3V
GND	GND

Buzzer Connection

Buzzer Pin	ESP32 Pin
Positive	GPIO15
Negative	GND

11. Complete Arduino Code

```
#include <SPI.h>
#include <MFRC522.h>
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

#define SS_PIN 5
#define RST_PIN 22
#define BUZZER 15

MFRC522 rfid(SS_PIN, RST_PIN);

float total = 0;

struct Product {
  String uid;
  String name;
  float price;
};

Product products[] = {
  {"D3A12B1C", "Milk", 35.0},
  {"F1A2456B", "Bread", 25.0},
  {"AABCCDD", "Rice", 60.0}
};

void setup() {
  Serial.begin(115200);
  SPI.begin();
  rfid.PCD_Init();

  pinMode(BUZZER, OUTPUT);

  if(!display.begin(SSD1306_SWITCHCAPVADDR, 0x3C)) {
    Serial.println("OLED Failed");
    while(1);
  }

  display.clearDisplay();
```

```

display.setTextSize(1);
display.setTextColor(WHITE);

display.setCursor(0,0);
display.println("AI Smart Cart");
display.println("Place Product");
display.display();
}

void loop() {

  if (!rfid.PICC_IsNewCardPresent())
    return;

  if (!rfid.PICC_ReadCardSerial())
    return;

  String content = "";

  for (byte i = 0; i < rfid.uid.size; i++) {
    content += String(rfid.uid.uidByte[i], HEX);
  }

  content.toUpperCase();

  Serial.println(content);

  bool found = false;

  for(int i=0; i<3; i++) {
    if(content == products[i].uid) {
      found = true;
      total += products[i].price;

      digitalWrite(BUZZER, HIGH);
      delay(200);
      digitalWrite(BUZZER, LOW);

      display.clearDisplay();
      display.setCursor(0,0);
      display.println("Product Added");
      display.println(products[i].name);
      display.print("Price: Rs ");
      display.println(products[i].price);
      display.print("Total: Rs ");
      display.println(total);
      display.display();
    }
  }
}

```

```

        Serial.println(products[i].name);
        Serial.println(products[i].price);
    }
}

if(!found) {
    display.clearDisplay();
    display.setCursor(0,0);
    display.println("Unknown Product");
    display.display();
}

rfid.PICC_HaltA();
}

```

12. AI Object Detection Python Code

```

import cv2

model = cv2.dnn.readNet("yolov3.weights", "yolov3.cfg")

classes = []
with open("coco.names", "r") as f:
    classes = [line.strip() for line in f.readlines()]

camera = cv2.VideoCapture(0)

while True:
    ret, frame = camera.read()

    height, width, channels = frame.shape

    blob = cv2.dnn.blobFromImage(frame, 0.00392, (416,416), (0,0,0), True,
crop=False)

    model.setInput(blob)

    output_layers = model.getUnconnectedOutLayersNames()
    detections = model.forward(output_layers)

    for output in detections:
        for detect in output:
            scores = detect[5:]
            class_id = scores.argmax()

```



```

        confidence = scores[class_id]

        if confidence > 0.5:
            center_x = int(detect[0] * width)
            center_y = int(detect[1] * height)
            w = int(detect[2] * width)
            h = int(detect[3] * height)

            x = int(center_x - w/2)
            y = int(center_y - h/2)

            label = str(classes[class_id])

            cv2.rectangle(frame, (x,y), (x+w,y+h), (0,255,0), 2)
            cv2.putText(frame, label, (x,y-10), cv2.FONT_HERSHEY_SIMPLEX,
0.5, (255,255,255), 2)

        cv2.imshow("AI Smart Cart", frame)

        key = cv2.waitKey(1)

        if key == 27:
            break

camera.release()
cv2.destroyAllWindows()

```

13. Algorithm

Step-by-Step Algorithm

1. Start system
 2. Initialize ESP32 and peripherals
 3. Scan RFID tag
 4. Identify product
 5. Display product details
 6. Update total bill
 7. Verify product using AI camera
 8. Check weight using load cell
 9. Alert if mismatch detected
 10. Send bill to cloud
 11. Stop
-

14. Advantages

- Saves customer time
 - Reduces manpower
 - Accurate billing
 - Prevents theft
 - Smart shopping experience
 - Easy inventory management
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15. Applications

- Supermarkets
 - Shopping malls
 - Smart retail stores
 - Automated billing systems
 - Cashier-less stores
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16. Future Enhancements

- QR code payment integration
 - Mobile app for live billing
 - AI recommendation engine
 - Voice-controlled cart
 - Autonomous robotic cart
 - Face recognition payment
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17. Expected Output

- Product automatically detected
 - Product name shown on display
 - Total bill updated instantly
 - AI verifies products
 - Theft alert generated on mismatch
 - Final bill stored online
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18. Viva Questions

Common Questions

1. Why ESP32 is used?
 2. What is RFID?
 3. Difference between barcode and RFID?
 4. Why load cell is used?
 5. What is object detection?
 6. What is IoT?
 7. Advantages of smart shopping cart?
 8. How theft detection works?
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19. Estimated Cost

Component	Approx Cost
ESP32	Rs 500
RFID Module	Rs 250
RFID Tags	Rs 150
OLED Display	Rs 300
Load Cell + HX711	Rs 400
ESP32-CAM	Rs 700
Buzzer and wires	Rs 100
Battery	Rs 300
Total	Rs 2700 approx

20. Conclusion

The AI Smart Shopping Cart is an innovative retail automation project that combines embedded systems, AI, IoT, and wireless communication to provide a faster and smarter shopping experience. This project reduces billing queues, improves efficiency, and provides a modern shopping solution suitable for future smart supermarkets.